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Long-Term Control and Toxicity After Iodine-125 Plaque for Ocular Melanomas

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Purpose/Objective(s): Ocular melanoma is a rare disease commonly treated with radiation therapy to preserve the eye. The purpose of this study was to describe local control and toxicity following Iodine-125 (I-125) plaque brachytherapy in a large, single-institution cohort of ocular melanoma.

Materials/Methods: All patients who received I-125 plaque for non-metastatic ocular melanoma at a single institution from 2003 to 2012 were reviewed. Patients were excluded if treated with proton beam radiation therapy, laser therapy, or transpupillary thermotherapy (TTT) prior to or in combination with I-125 brachytherapy. Patients with detailed ophthalmic exams prior to treatment and post-treatment exams at 12 and 24 months were analyzed for the following: change in visual acuity (VA), cataract surgery, vitreous hemorrhage (VH), radiation retinopathy (RR), and enucleation. Local control (defined as lesion stable or regressed) and additional treatments delivered for ocular melanoma were also recorded.

Results: A total of 165 patients were treated. Of these, 107 had complete ophthalmic exams at baseline, 12 months, and 24 months. All patients had minimum follow up of 24 months. Median patient age at the time of I-125 was 66 years (range 39-93). Median tumor height was 4.88 mm (range 2-10.9 mm) and median plaque diameter was 16 mm (range 10-20 mm). The majority of patients (86%) were prescribed 80 Gy to tumor apex or 5 mm from sclera (if height less than 5 mm). The remainder received 65-75 Gy due to normal tissue constraints (sclera dose less than or equal to 450 Gy) and these all had tumor height of 8 mm or greater. The mean dose delivered to the apex and sclera was 96 Gy and 336 Gy, respectively. Local control at 24 months was 95%. The mean tumor height of those that failed was 8.2 mm. Secondary treatment for ocular melanoma was required in 24% of patients. Toxicity is shown in the table below. Visual acuity declined after 12 months of follow up and was significantly lower at 24 months when compared to baseline VA. Thirteen patients required enucleation, 9 (69%) for recurrence and 4 (31%) for toxicity. At 24 months, 11 patients (10.3%) required surgery for cataracts, 48 (45%) developed RR, and 15 (14%) developed VH. Of the patients who developed RR or VH, 15 (31%) and 4 (26.7%) required treatment, respectively.

Abstract 2822; Table 1.

	Pretreatment (n = 107)	12 month (n = 107)	24 month (n = 107)
Median VA	20/40	20/45 (P = 0.11, baseline)	20/70 (P = 0.002, baseline)
VA 20/50 or better	51.4%	55.1%	37.4%
VA 20/200 or better	86.0%	77.6%	62.6%
VH	4.7%	2.8%	14.0%
RR	n/a	17.8%	44.9%
Enucleation	n/a	5.6%	12.4%

Conclusion: Plaque brachytherapy is a safe and effective treatment for ocular melanoma. Local control is excellent at 2 years with low overall toxicity rates. However, greater tumor height was associated with a decrease in local control. There was no acute or dramatic change in VA, but VA continues to decline over the follow-up period with most patients maintaining functional vision.

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Nontarget Swallowing Muscles Dose-Volume Correlates of Patient-Reported Outcomes After Oropharyngeal Intensity Modulated Radiation Therapy (IMRT)

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Purpose/Objective(s): Development of Normal Tissue Control Probability (NTCP) model(s) that include robust normal tissue endpoints, including patient-reported outcomes (PRO), is highly needed. We aim to validate the association between chronic swallowing and choking items in the MD Anderson Symptom Inventory-Head and Neck module (MDASI-HN) and dosimetric parameters of non-target swallowing muscles after IMRT of oropharyngeal cancer (OPC).

Materials/Methods: OPC patients receiving IMRT and/or systemic therapy in our institution between 2001 and 2013 who completed the MDASI-HN module >12 months post-IMRT completion were reviewed. Swallowing and choking symptoms were coded as a binary outcome comparing patients who reported moderate to severe dysfunction (MDASI-HN scores > 4 on a 10 point scale) to patients with mild or no dysfunction (scores ≤ 4), as well as comparing any symptoms scores (≥ 1) to asymptomatic scores (0). IMRT plan data were extracted from the treatment planning system and auto-segmented using a custom ROI library. Segmented ROIs included: pharyngeal constrictors, pterygoids, digastrics, tongue, mylo/geniohyoid, genioglossus, and soft palate. ROI-specific dose-volume histograms (DVHs) were calculated. Recursive partitioning analysis (RPA) was used to identify dose-volume effects associated with binary MDASI-items.

Results: Of the 477 reviewed patients, 230 were excluded due to lack of retrievable IMRT plans or MDASI-HN completion, leaving 247 for analysis. Median age was 55 years (range 30-85). Tumors originated from tonsillar fossa in 138 patients (55%) and base of tongue in 96 (39%). 220 patients (89%) were AJCC stage III-IV. Mean prescribed dose was 66.7 Gy (SD 2.6). 85 patients (34%) reported moderate to severe swallowing dysfunction and 143 (58%) reported swallowing score ≥ 1. 46 patients (19%) reported moderate to severe choking dysfunction and 146 (60%) reported choking score ≥ 1. 43 patients (17%) reported moderate to severe scores in both items and 117 (47%) reported ≥ 1 score in both items. RPA showed no association between single item swallowing and choking binary splits and dosimetric parameters of non-target ROI. However, RPA found DVH-derived genioglossus V65 (volume receiving 65 Gy) and middle pharyngeal constrictor V65 were significantly associated with patients reporting moderate to severe scores in both items — swallowing and choking (RPA logworth $p < 0.02$; cumulative ROC AUC 0.76; chi-square $P < 0.01$).

Conclusion: DVH parameters in our set were not sensitive discriminate single item patient-reported swallowing or choking items scores, but significantly predicted moderate to severe scores in composite swallowing-related symptoms. DVH-derived correlates of PROs were distinct from those previously reported using clinician-graded chronic radiation-associated dysphagia endpoints. Further efforts are needed to define parsimonious models combining PRO and clinical endpoints for robust clinically usable dosimetric constraints in future NTCP models.

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